Code No: 115DT **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, August - 2022
COMPUTER NETWORKS
(Common to CSE, IT)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) What is the importance of layered architecture in network models? Discuss in detail.
b) Explain about TCP/IP reference model with a neat sketch. [7+8]

2.a) What are the design issues of Data Link Layer? Explain in brief.
b) Write about twisted pair cables and coaxial cables with neat diagram. [7+8]

3.a) Compare and contrast CSMA/CD and CSMA/CA for channel allocation.
b) Explain about spanning Tree Bridge. [8+7]

4.a) Discuss about Network hardware components in detail.
b) Write about Ethernet Mac sub layer in detail. [7+8]

5.a) Explain in detail about design issues of network layer.
b) Explain about the principle of optimality. [10+5]

6.a) Write about count to infinity problem.
b) How to achieve quality of service using leaky bucket algorithm? [7+8]

7.a) Discuss about ARP and RARP protocols in detail.
b) What are the advantages of tunneling? [10+5]

8.a) Explain about the TCP connection release.
b) What is DNS? Explain the architecture of DNS Servers in Internet. [7+8]

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